Probability calculations

1 In a class, 5 students play both football and tennis, 13 students in total play tennis, and 11 in total play football.

a How many students only play football?

b How many students play only one sport?

c If a random student is chosen from the group, find the probability that the student only plays tennis.

2 Consider the following list of numbers:

$$1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144$$

If a number from the list is chosen at random, find:

a The probability that it is an even number that doesn't contain a 3.

b The probability that it contains a 3 or is an odd number.

c The probability that it is either odd or contains a 3, but not both.

3 A grade of 234 students are to choose to study either Mandarin or Spanish (or both). 134 students choose Mandarin and 120 students choose Spanish.

a How many students have chosen both languages?

b If a student is picked at random, find the probability that the student has chosen Spanish only.

c If a student is picked at random, find the probability that the student has not chosen Mandarin.

4 In a music school of 129 students, 83 students play the piano, 80 students play the guitar and 14 students play neither. Find the probability that a student chosen at random plays:

a Both the piano and the guitar.

b The piano or the guitar.

c Neither the piano nor the guitar.

5 In a survey of 31 students, it was found that 16 students play tennis and 14 students play hockey. 2 students play none of these sports, 8 play both tennis and cricket, 7 play both cricket and hockey, 6 play both tennis and hockey and 3 play all three.

Find the probability that a randomly selected student plays all three sports.

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6 In a study, some people were asked whether they were musicians or not. 25 responders said they were a musician, of which 10 were children. 25 children said they were not musicians, and 13 adults said they are not musicians.

a How many people were in the study?

b State the proportion of responders that are musicians.

c State the proportion of adults that are musicians.

7 In a particular high school where there are 91 year 8 students, 40 students study History only, 36 students study French only and 10 students study both. If a student is randomly chosen, find the probability that this student is studying:

a History and French.

b History only.

c French only.

d French or History.

e Neither subject.

8 A student is making a Venn diagram about politicians in the last two elections. Looking at a group of 24 politicians, 14 ran in the first election and 19 ran in the second election.

a Given that every politician examined was in at least one election, how many politicians ran in both elections?

b If a politician is randomly chosen in the second election, find the probability they were also in the first election.

9 In a survey, 104 students were asked if they are left or right-handed, of which 49 were female. 50 male students said they were right-handed, and 10 female said they are left-handed.

a How many male students were in the study?

b State the proportion of students that are left-handed.

c State the proportion of male students that are left-handed.

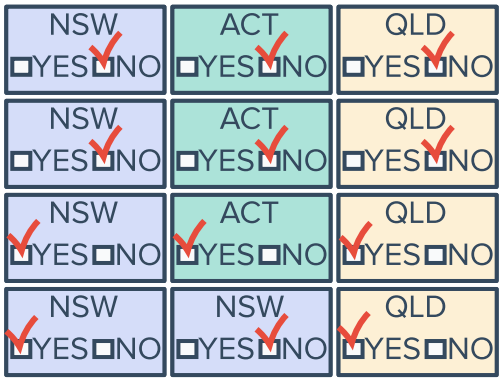
10 A small magazine asked people from different states to send in a vote on whether they supported Daylight Saving Time. The diagram shows the proportion of people that voted YES or NO in NSW, ACT and QLD:

a State the proportion of people who voted "YES".

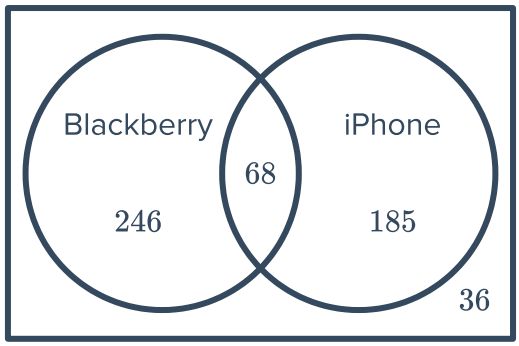
b State the proportion of "NO" votes that are from NSW.

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c Considering just the voters from NSW and ACT, state the proportion of the votes that are NSW "YES" votes.



11 The Venn diagram shows the decisions of 535 consumers choosing to buy an iPhone and consumers choosing to buy a Blackberry:



If a consumer is selected at random, find the probability that he chose to buy:

a A Blackberry.

b A Blackberry only.

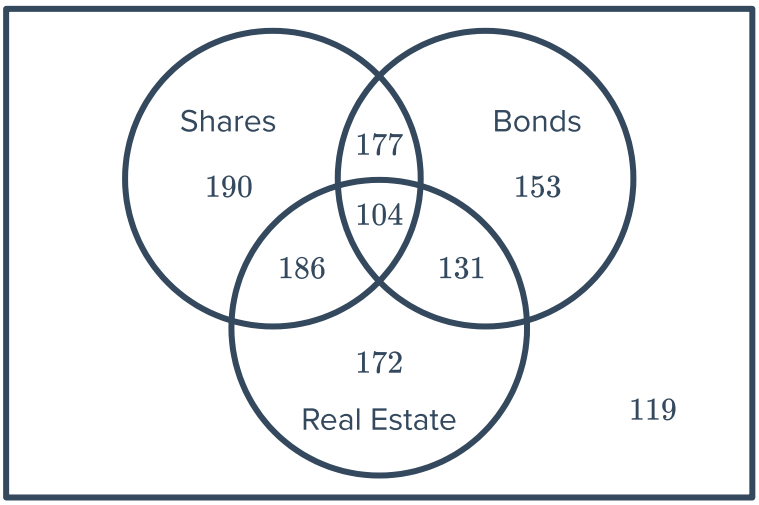
c Both phones.

d An iPhone or a Blackberry.

e Neither phones.

f An iPhone but not a Blackberry.

12 The Venn diagram depicts the investment choices of 1232 investors:



Find the probability that an investor randomly selected has investments in:

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a Bonds

b Bonds and real estate.

c Bonds or real estate.

d Bonds and real estate but not shares.

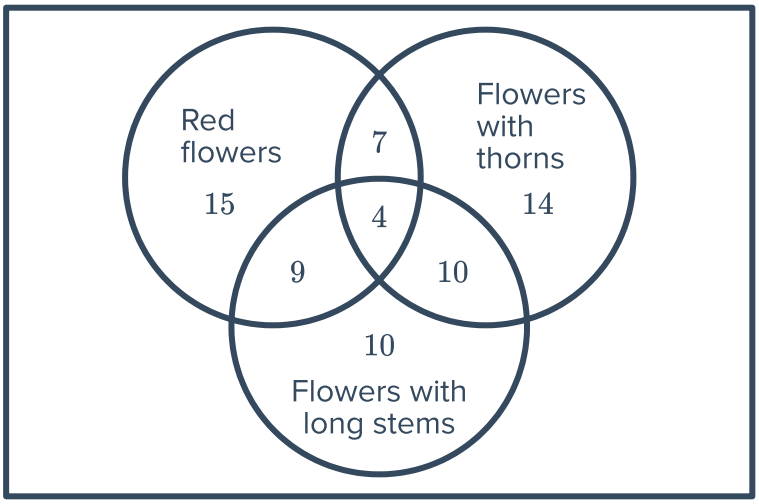
e Shares, bonds and real estate.

f Real estate and shares but not bonds.

g Shares, bonds or real estate.

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13 A florist collected a sample of her flowers and divided them into the appropriate categories. as shown in the Venn diagram:



Find the probability that a flower is:

a Not red but has thorns.

b Not red and does not have thorns.

14 In a group of 183 primary and senior students, 96 are primary students. The students fell into three categories of travel to school - by bus, car, or walking. 113 students get to school by bus, 60 are primary students. 48 students get to school by car, 25 are primary students.

a How many senior students catch the bus to school?

b How many senior students are there in total?

c Find the probability that a senior student walked to school.

15 Sophia asked some people in her community whether they were vegetarian or not. 29 responders said they were vegetarian, of which 8 were children. 14 children said they were not vegetarian, and 11 adults said they are not vegetarians.

a What proportion of responders are vegetarian?

b What proportion of adults are vegetarian?

16 99 students were asked if they choose to study on the night before an exam or choose to party. 51 students choose to party while 59 choose to study. 10 neither choose to study nor party.

a Find the number of students that chose to study and party

b Find the probability that a student chosen at random chose:

i Not to party.

ii To study and party.

iii To study only.

iv To study or party.

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17 The employees of Pentagonal Plumbing were discussing where they should hold their end of year party. Of all of the employees:

- 53 would go to a restaurant
- 67 would go to a bowling alley
- 72 would go to a theme park
- 16 would only go to either a restaurant or a bowling alley
- 27 would only go to either a bowling alley or the theme park
- 17 would only go to either a restaurant or the theme park
- 8 would not go to a restaurant or a bowling alley or to the theme park
- 6 would go to all three places.

a How many people in total work at Pentagonal Plumbing?

b Given that the restaurant isn't available, state the proportion of the employees who still have another option.

18 Out of 100 students in a school, there are 55 students who are taking Physics classes, 50 students who are taking Chemistry classes and 15 students who are not enrolled in Physics or Chemistry classes.

Find the probability that the student is:

a Taking Physics classes only.

b Taking Chemistry classes only.

c Taking both Physics and Chemistry classes.

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